

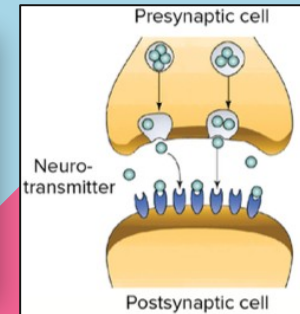
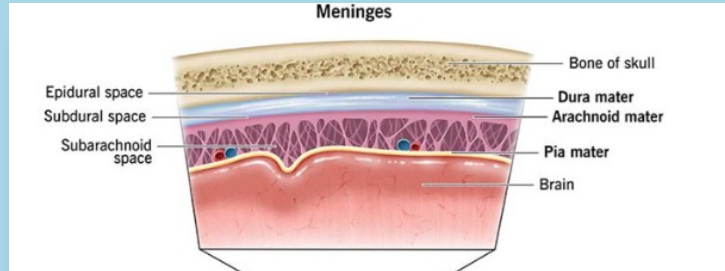
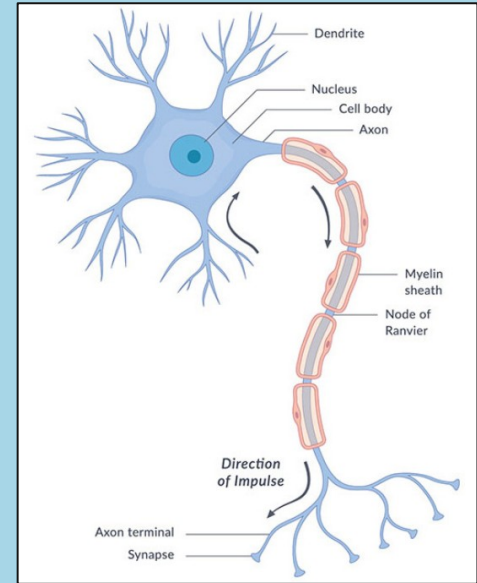


Science Review

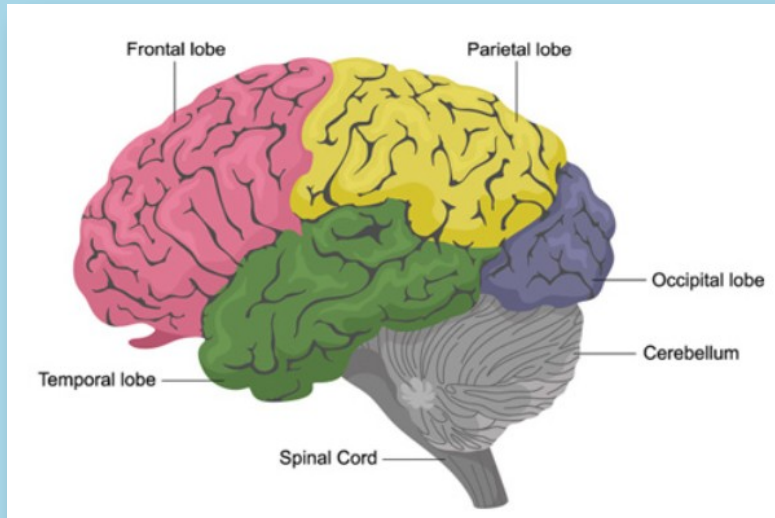
By Matija Benko and Connor Larson

Cells and the Anatomy of the Brain

- **Neuron**: basic building block of the nervous system
 - A neuron's structure determines its function
 - **Neurotransmitters**: chemicals that trigger a neuron to fire
- **Grey matter**:
 - Outer part
 - Mostly dendrites & soma
- **White matter**:
 - Inner part
 - Mostly myelinated axons
- **Microglia**: Immune cells of the brain
- **Meninges**:
 1. Dura Mater
 2. Arachnoid mater
 3. Pia Mater

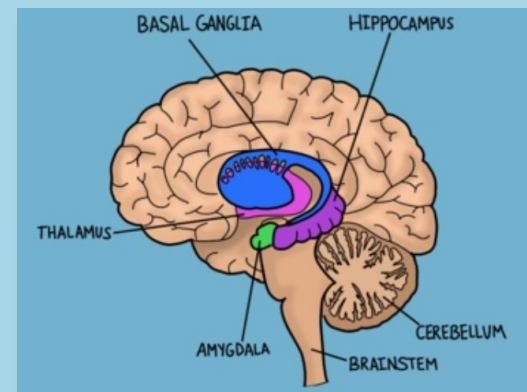


- LOBES:
 1. Frontal lobe - Rational thinking, planning
 2. Parietal lobe - body sensations & position
 3. Occipital lobe - Sight
 4. Temporal lobe - Hearing, music, rhythm, memory
- Cerebellum - motor system, learning smooth & coordinated movements



- SUBCORTICAL STRUCTURES:

1. Amygdala “almond” - addiction & anxiety
2. Hippocampus - long-term mem. & navigation
 - a. Damage -> no more long-term memories
3. Thalamus - comm. between body & brain
4. Hypothalamus - control pituitary gland & homeostasis(body’s balance)
5. Pituitary gland - releases growth hormones
6. Basal ganglia - control of planned movement

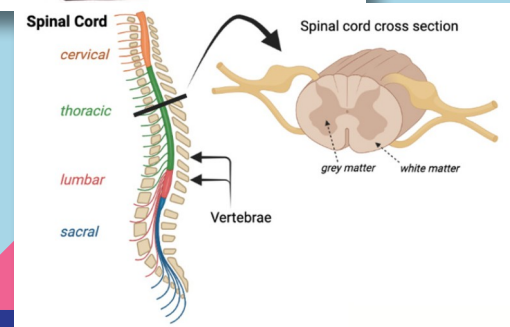
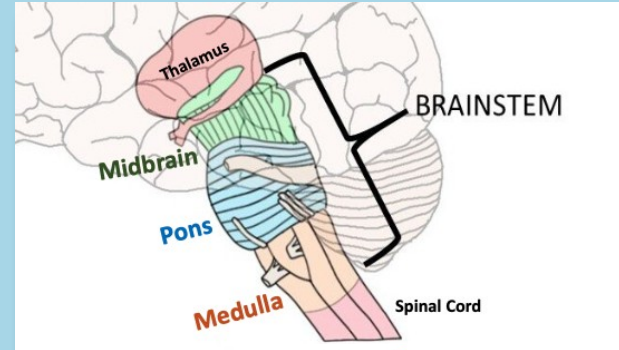


- **BRAINSTEM:**

1. Midbrain - eye movement
2. Pons - facial movements
3. Medulla - touch from face, swallowing, vomiting, regulating blood pressure, heart rate, breathing, ect.

- **SPINAL CORD:**

- Encased in vertebrae
 - Conducts information between the brain & body
1. Cervical - arms, neck, shoulders
 2. Thoracic - chest & torso
 3. Lumbar - hips & front legs
 4. Sacral - everything else

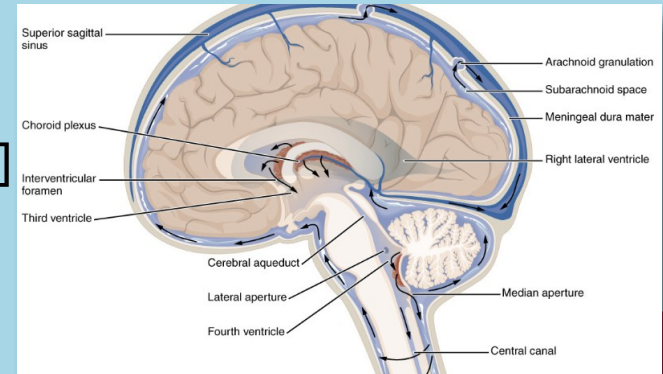


VENTRICULAR SYSTEM:

- Holes filled w/ Cerebrospinal fluid(CSF)
 - CSF helps Brain float, protects it & removes old signaling chemicals, toxins & infectious particles
- Each hemisphere has a lateral ventricle(near thalamus)
- Choroid Plexus lines each ventricle & makes new CSF

PERIPHERAL NERVOUS SYSTEM:

- Consists of Ganglia[comm. between body & brain]
- Easier to damage, but easier to regenerate
- more open to infection & toxins



Somatic Nervous System:

- info from & to tissues[Voluntary]

Autonomic Nervous System:

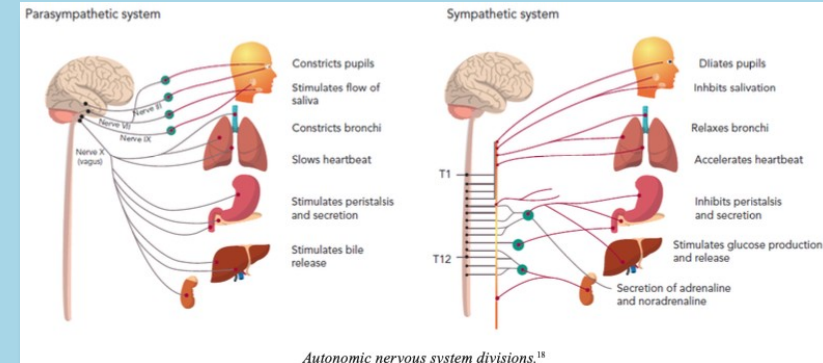
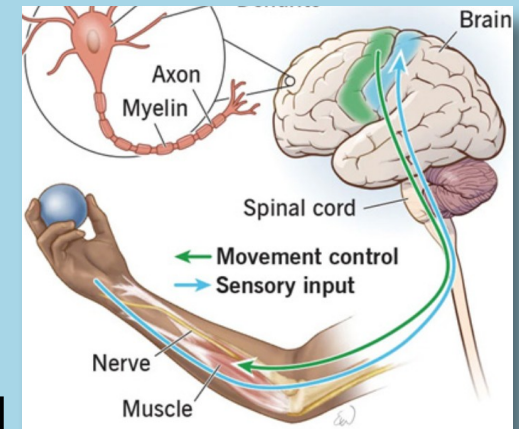
- regulates inner organs & smooth muscle[Involuntary]
- Both divisions target the same organs & are inhibitory

1. Sympathetic - “Fight-or-flight”

- a. found in thoracic & lumbar spinal cord

2. Parasympathetic - “rest and Digest”

- a. Found in brainstem & the sacral spinal cord
- b. Slower than Sympathetic
- c. They do the opposite of Sympathetic





Neural Communication

(This is a long one, I'm so sorry)

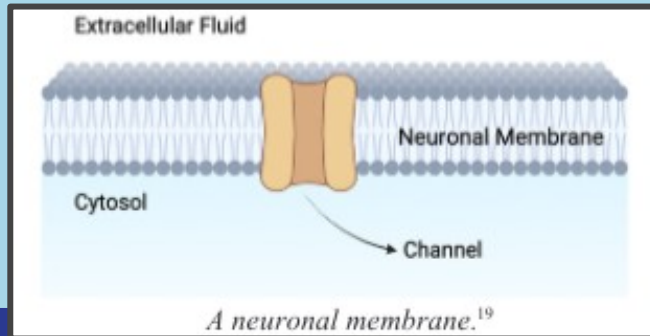
Neurons can also send *electrical signals* via ions. These electrical states caused by ions are **membrane potentials**.

THE RESTING MEMBRANE POTENTIAL

- measured in **millivolts (mV)**
- resting potential (off) & the action potential (on)
- Dictated by the cytosol (fluid inside the cell), **extracellular fluid**, & the neuronal membrane separating them.

The neuronal membrane is a selective phospholipid bilayer (think cell wall). It allows the amount/concentration of ions within the fluids to differ inside versus outside the cell due to its selectivity

The ions must find a **channel** (opening) that's **permeable** (accessible) to that kind of ion to pass from one side to the other.



Passive transport, an ion moves to another naturally. **Ionic concentration** (or diffusion) & **electrical charge** (or potential).

- **Diffuse:** movements of ions from a higher concentration to lower concentration.
- **Electrical potential:** Ions in the extracellular fluid & inside the cell behave like magnets.

Ion channels: gates in the neuronal membrane particular ions can pass through

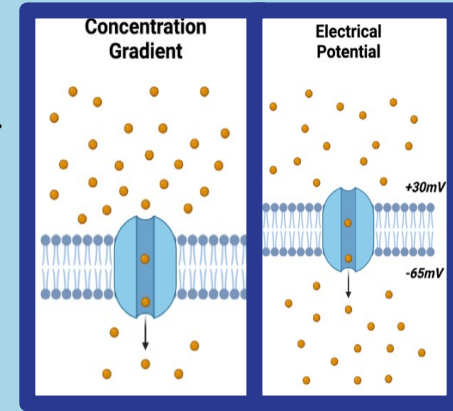
- **Voltage-gated:** Membrane
- **Ligand gated:** Particular m

Active transport, an ion goes aga

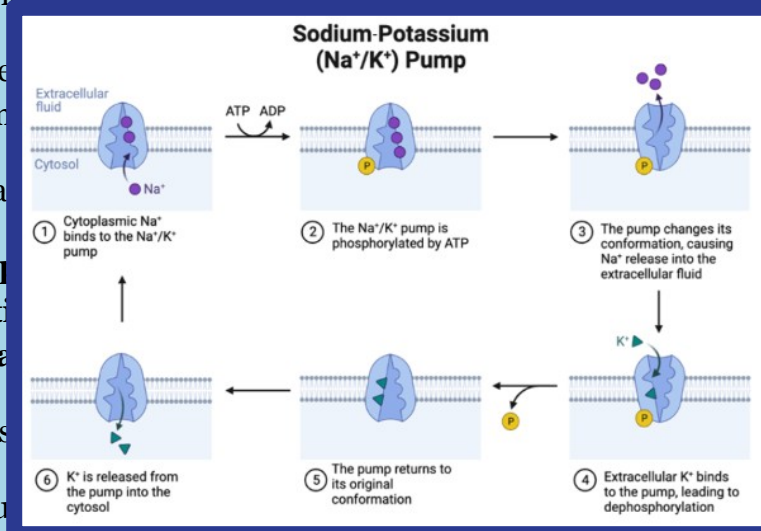
- **Sodium-potassium pump**
- Changes shape/**conformat**
- **potassium leak cha**

When a neuron is at rest, the potass

- The potential membrane cau
- cell (the **electrical driving force**) repels other K^+ ions into the cell via an open channel. When equal w/ the **chemical driving force** (result of initial concentration gradient causing ions to diffuse out of the cell), there is no net movement of ions, for every K ion that leaves, another enters.



energy source in our cells).
against concentration gradients.



ositive charge built up outside the

The membrane potential at which a particular ion reaches equilibrium is its **equilibrium potential**. Electrical potential & the chemical gradient are proportional to the difference between the inside & outside of the cell.

- equilibrium potential for a given ion is calculable via **the Nernst equation**.
- **Goldman equation:** factors in the relative permeability of the cell to multiple different ions.

> E_{Cell} = cell potential
 > E^0 = Cell potential under standard conditions
 > R = Universal gas constant ($8.314 \frac{\text{J}}{\text{Kmol}}$)
 > T = Temperature
 > n = Number of electrons transferred in the reaction
 > F = Faraday constant ($96485 \frac{\text{C}}{\text{mol}}$)
 > Q = Reaction Quotient

$$E_{\text{Cell}} = E^0 - \left(\frac{RT}{nF} \right) \ln Q$$

$$V_m \approx -60 \text{ mV} \log_{10} \frac{P_K[K]_{\text{in}} + P_{Na}[Na]_{\text{in}} + P_{Cl}[Cl]_{\text{out}}}{P_K[K]_{\text{out}} + P_{Na}[Na]_{\text{out}} + P_{Cl}[Cl]_{\text{in}}}$$

WHAT EACH FIGURE MEANS:

[Link Here](#)

THE ACTION POTENTIAL

Before & after an **action potential** (electrical potential across the neuronal membrane when the cell is active & firing), a neuron is at its:

- **Rising phase:** membrane potential = increasingly positive due to cations (**depolarization**).
 - The membrane potential continues becoming positive, usually until $\sim +30$ to $+40$ mV. (**overshoot**).
- **Falling phase:** membrane potential = increasingly negative due to cations leaving (**repolarization/hyperpolarization**).
 - Continues past resting potential until ~ -80 mV (**undershoot/ afterhyperpolarization**).

Most **stimuli** work to reach a membrane potential **threshold** (-65 to -50 mV), voltage-gated sodium channels open, beginning the rising phase. Once the threshold is reached, the process cannot stop (the **all-or-none principle**). (like pushing a button, you either do it or you don't)

Voltage-Gated Sodium Channels = very efficient in rapidly depolarizing the cell (electrical & chemical driving forces). Potential peaks at +30 to +40 mV, then the channels become blocked/inactivated (can't open regardless of surrounding conditions). This is the **absolute refractory period**, during which no other action potential can start.

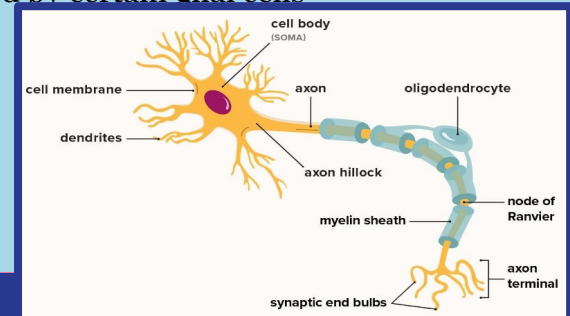
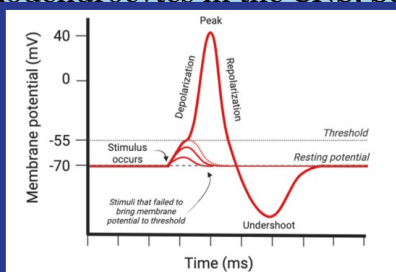
Voltage-Gated Potassium Channels open at the peak & do the exact opposite, causing K^+ ions to leave (**efflux**) rapidly & remove a + charge from the inside of the cell, hyperpolarizing it. These channels are delayed in opening & closing (causing undershoot). Thus, it is called a **delayed rectifier**.

- Due to the undershoot, it is possible, but hard, to start another action potential(**relative refractory period**).

Action potentials travel (**propagate**) down neuron's axon rapidly by consecutive action potentials. Otherwise, the signal would fade away (think lighting a fuse).

- Propagation of the action potential along the axon only occurs in one direction—usually from the soma/axon hillock to the axon terminals at the end.

To speed things up, large sections of the axon are insulated w/ **myelin**, a sheath formed by certain glial cells (oligodendrocytes in the CNS, Schwann cells in the PNS), like copper wires w/ rubber.



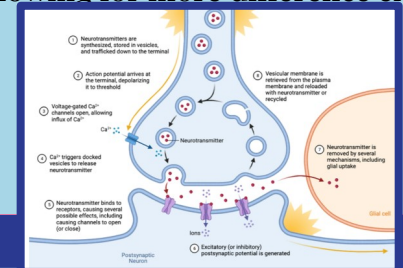
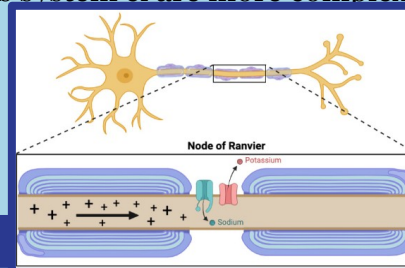
The action potential is regenerated at the **Nodes of Ranvier**, gaps in the myelin where the axon contact extracellular fluid. Voltage-gated Na & K channels are clustered at these nodes. Conduction down a myelinated neuron is called **saltatory conduction**, (Latin (*saltare*) meaning “to jump”) as the message jumps down the inside of the axon, node to node, until it reaches the terminal.

SYNAPTIC TRANSMISSION

Once the action potential reaches the terminal, **synaptic transmission** occurs. The presynaptic neuron = neuron releasing a signal at the synapse, the postsynaptic neuron = the receiver. The axon terminal of the presynaptic cell contains many **mitochondria** to make the ATP that powers many aspects synaptic transmission.

The structure of the axon is made by long cytoskeletal elements, **microtubules**, which disappear at the axon terminal. There are 2 main types of synapses in the CNS: **electrical synapses** & **chemical synapses**.

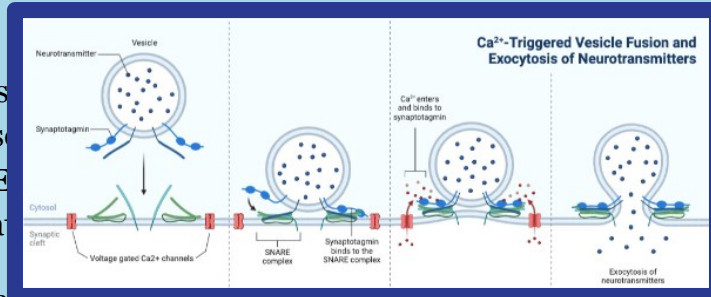
- Electrical synapses: The 2 neurons have a single set of channels (**gap junctions**) touching their neuronal membranes. Thus, the cations forming the action potential can flow directly from cell to cell. But the signal cannot be altered in any way so it is a bit limiting & uncommon.
- Chemical synapses: Make up most synapses in the nervous system & are more complex, allowing for more difference & can make more complex neuronal networks.



The presynaptic cell takes the action potential, an electrical signal, & translates it into a chemical signal (electrochemical process) that it releases & sends to another neuron. This transformation takes place in the axon terminal.

Once reached, voltage-gated calcium (Ca^{2+}) channels open. The ions rapidly enter the cell. Calcium then plays a critical role in initiating the release of chemicals, neurotransmitters (NT for this presentation & my sanity), to carry the signal across the synaptic gap/cleft.

- NTs are generally produced in the soma & sent to the axon terminal for later use. They are released from the presynaptic membrane-bound proteins (**SNARE** proteins) in the presynaptic membrane where all the activity related to neurotransmission occurs.



Vesicles (small spheres of membrane), containing the neurotransmitters, crossing the synaptic gap via special proteins. The region of the neuronal membrane where this occurs is the synaptic terminal.

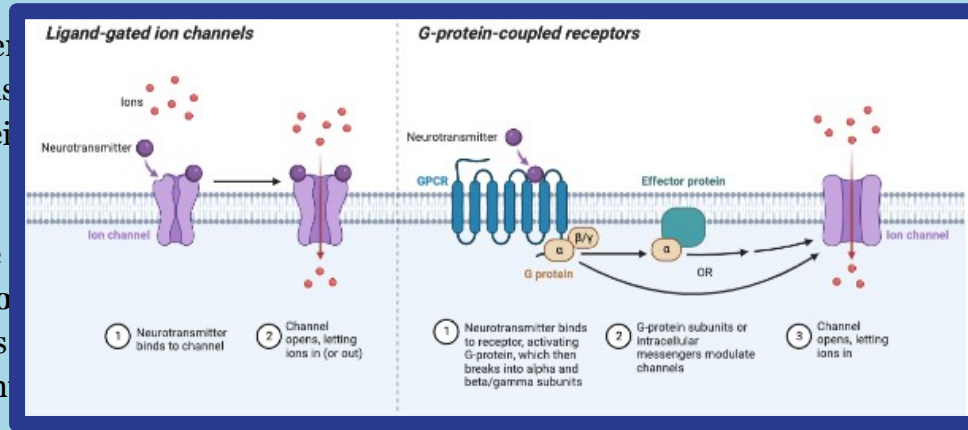
Upon triggering release, influxing Ca^{2+} (near the vesicles) binds to Ca^{2+} for the SNARE proteins, **synaptotagmin**, which changes its structure, bringing the vesicle up to the neuronal membrane so that the membranes fuse. This allows a pore so the NT molecules diffuse into the synaptic cleft & float toward the postsynaptic cell. This is **exocytosis**. The vesicle membrane is then recovered & separated back out from the larger neuronal membrane (via **endocytosis**) to ensure overall neuronal membrane stays a consistent size.

The NT itself must clear out for the process to repeat. This can happen via enzymatic degradation, uptake by glial cells, /simply letting the NT to diffuse over time. Some terminals feature **reuptake** transporter proteins, which reabsorbs the NTs it dispersed for later use.

Once released, NTs diffuse across the synaptic cleft & bind to receptors embedded in the membrane of the other cell. These receptors are often specific for 1 type of NT (like a key to a lock). The area of the postsynaptic membrane by the synaptic cleft is so thickly clustered w/ receptors & their corresponding parts that this area is called the **postsynaptic density**. The 2 main categories of postsynaptic receptors are ligand-gated ion channel receptors (**ionotropic receptors**) & **G-protein coupled receptors (metabotropic receptors)**.

Ionotropic receptors are membrane proteins that have specific binding sites on their extracellular side. However, these ion channels are also embedded in the membrane. When a ligand is bound, the channel opens, allowing ions to flow across the membrane.

Ions that hyperpolarize the membrane are said to have an **inhibitory** effect. Ions that depolarize the membrane are said to have an **excitatory** effect. Excitatory cations flowing through this channel might result in a brief hyperpolarization that might



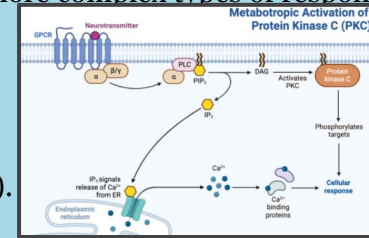
in a ligand is bound. NT molecules are bound to the receptor, which results in the movement of ions across the membrane. This might depolarize the membrane, which might result from a brief hyperpolarization. Similarly, the brief

One EPSP/IPSP on its own is small & brief, so it cannot do anything all by itself. Instead, the EPSPs & IPSPs coming in from all the synapses are summed together, a phenomenon known as summation (**synaptic integration**). Whether neurons fire or not is the result of the combination of messages they receive.

Spatial summation: when multiple EPSPs/IPSPs occur simultaneously at different synapses at different spatial locations across the dendrites of a neuron & are summed. **Temporal summation:** when multiple EPSPs (or IPSPs) occur in rapid succession at the same synapse & are summed up over time. The axon hillock is the point where the inputs are summed.

Metabotropic receptors (G-protein-coupled receptors ,GPCRs) work slower, but allow for more complex types of responses. (I'm going to call G Proteins GPs from now on).

- Activated via the binding of NT just like in the ionotropic receptors.
- Activate GPs when NT binds.
- GPs divide into subunits, each activate their set of signal proteins (effector proteins).



Effector proteins depend on neurons & synapses. One common pathway is for the GP to bind to & open a special ligand-gated ion channel, letting ions flow & the ion channel is the effector protein. There are also ligand-gated ion channels open by binding to other effector proteins that have been activated by GP subunits.

Some metabotropic receptors can have widespread effects all over the postsynaptic cell. Many of these pathways utilize cascades of proteins “downstream” from the GP, which alters cell metabolism, (hence “metabotropic” receptors). Some such pathways branch, targeting several different types of effector proteins, each w/ its own role.

- Example: metabotropic activation of protein kinase C (PKC), which activates many important molecules by adding a phosphate group to (**phosphorylate**) them, the main role of protein kinases. Here, the GP subunits the NT triggers by binding to the GPCR bind to the membrane-bound effector protein **phospholipase C (PLC)**, splitting it in two: **inositol-1,4,5-triphosphate (IP_3)** & **diacylglycerol (DAG)**.

In the PKC example, each molecule is a **second messenger** (impacts other areas of cell function).

- In that case, IP₃ binds to IP₃-gated calcium channels within the membranes of the endoplasmic reticulum, allowing internal stores of calcium to flow into the cytosol, which then binds to many signaling molecules, activating them. Meanwhile, DAG stays bound to the membrane & activates another membrane-bound protein, PKC. Like other kinases, it phosphorylates target proteins, leading to numerous downstream effects, impacting the function of the entire cell.

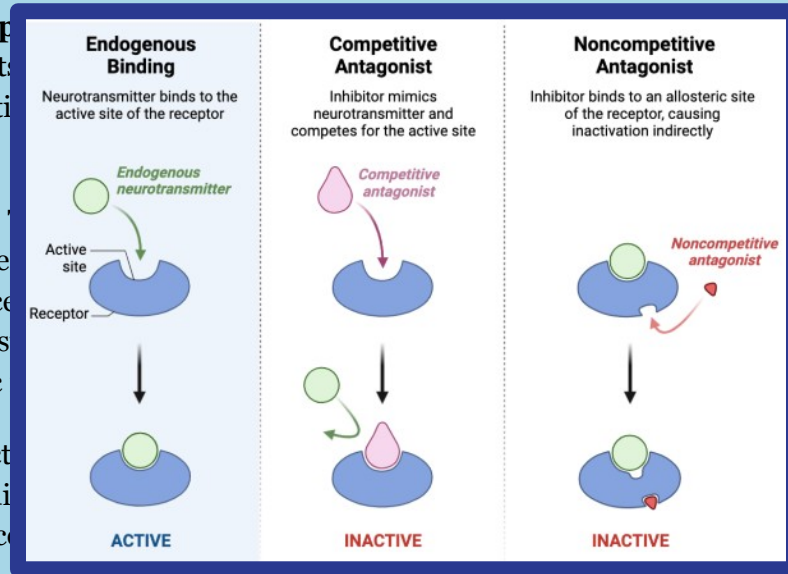
These **signal cascades** take longer to produce an effect than the simpler ionotropic pathways, but the result is **signal amplification**.

NEUROTRANSMITTERS

Some drugs (**serotonin selective reuptake inhibitors**) give the NT more time to bind & rebound to more postsynaptic receptors, & its behavior work by altering the function of the NT as an **agonist/antagonist**.

Agonist = enhance the activity of an NT. **Competitive agonists**, which compete for the binding sites on the postsynaptic receptor, blocking its binding site, making the existing NT's physiological effects on the postsynaptic cell.

Antagonist = decrease/fully block the activity of an NT by blocking the binding site as a **competitive antagonist**,/indirectly by blocking the NT (**allosteric site**) altering the receptor's effect.



of **serotonin**, a NT, giving the NT more time to bind & rebound to more postsynaptic receptors, & its behavior work by altering the function of the NT as an **agonist/antagonist**.

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Drugs that interfere w/ NT synthesis act as antagonists, & drugs that stimulate this process act as agonists.

NTs are either excitatory, inhibitory, or **modulatory** (have more complex functions besides affecting membrane potential) Generally, each neuron produces 1 type of NT. Neurons are named based on what type of NT they produce (dopamine neuron = **dopaminergic** neuron.)

- **Glutamate** = most common excitatory NT. Plays a role in functions like learning & memory. Excess glutamate release is associated w/ seizures & sometimes cell death/neuronal death (**excitotoxicity**) due to prolonged (+) charge.
- **Gamma-aminobutyric acid (GABA)** = most common inhibitory NT. Known for its calming effect. Problems w/ GABA signaling are associated w/ mood disorders, sleep & motor problems, etc. Like glutamate, too little GABA can cause excitotoxicity. Glutamate & GABA work like an on & off switch in the brain, forming a delicate balance.

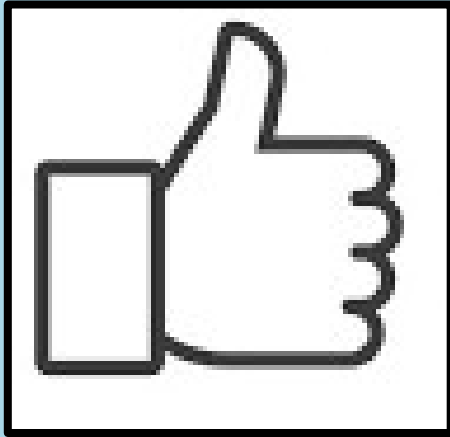
Catecholamines = modulatory NTs made from the same precursor molecule, **tyrosine**. Includes **dopamine**, **norepinephrine**, & **epinephrine** (or noradrenaline & adrenaline). Neurons producing these are found in parts of the brain responsible for movement, mood, attention, & the function of the internal organs.

- Dopamine comes from the **ventral tegmental area (VTA)** & the **substantia nigra** of the basal ganglia. Dopamine from VTA is important for regulating reward system/motivation (Cause for addiction). Dopamine from the substantia nigra plays an important role in initiating voluntary movement. (Death of these cells is seen in Parkinson's Disease).
- Norepinephrine is produced from dopamine via a special enzyme found bundles of neurons (or nuclei) in the brain, the main one being the **locus coeruleus** (Latin for "blue spot"), found in the brainstem. Norepinephrine-producing neurons send axons all over the brain to regulate mood & attention (and many other functions).
- Epinephrine (adrenaline) can be used to treat many problems, including clot formation & restarting the heart in the case of cardiac arrest. Produced by the adrenal glands & a small number of neurons in the brainstem, where it plays a key role in mediating the "fight or flight" response via stimulation of the SNS.

Serotonin (or **5-hydroxytryptamine (5-HT)**) is produced by a relatively small number of neurons w/ cell bodies found in the raphe nuclei of the brainstem. A significant proportion of it targets the digestive tract, but in the brain, it plays a role in numerous functions, most famously mood & emotion. Thus, it's a commission target for drugs used to treat anxiety & depression. (Drugs include hallucinogens such as LSD, MDMA, & psilocybin (the active ingredient in “magic mushrooms”). Many of these are highly efficient serotonin agonists.

Acetylcholine = primary NT found at the **neuromuscular junction** (where neurons synapse onto the muscles, informing muscles down how to move & contract.) Due to the role of acetylcholine at the neuromuscular junction, many important drugs target the acetylcholine system. Large doses of some acetylcholine antagonists can stop the heart, but small doses can potentially treat heart conditions/other problems. Nicotine, acts as a partial acetylcholine agonist as well

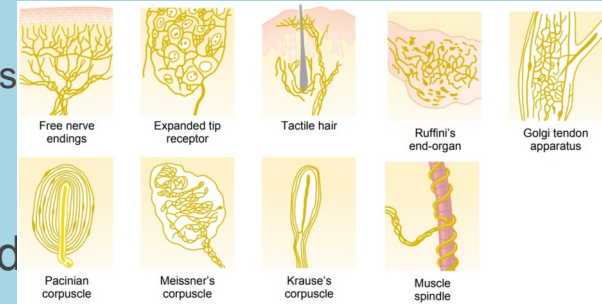
- **(THAT'S ANOTHER REASON WHY NICOTINE IS BAD FOR YOU, DON'T SMOKE!)**



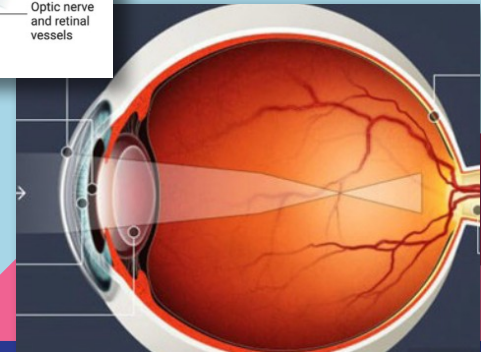
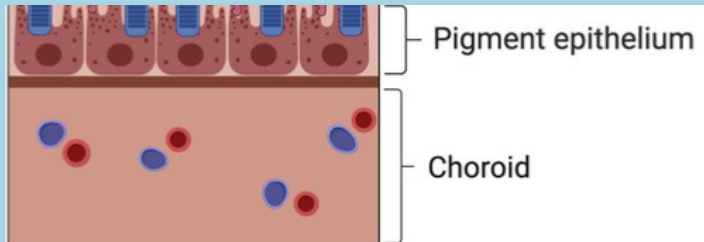
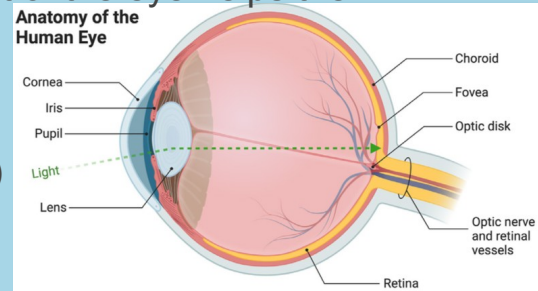
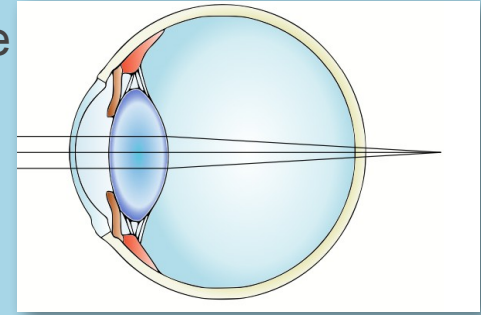


The Sensory and Motor Systems

- The 3 senses/systems convert stimuli into signals the NS can understand(sensory transduction),
 - **Somatosensation:** Any sensation involving physical contact(temp., pain, pressure, & vibration)
 - **Vestibular system:** sense of balance & spatial awareness
 - **Proprioception:** sense of where your organs are located
- most of these also have a sensory receptor
- **receptive field:** place where stimulus is detected
- You are not aware of all stimuli
- When a receptor is activated, it depolarizes, then an action potential(AP)/transmitter release
 - Photoreceptors are the exception
- **Parallel processing:** stimulus features are kept in separate “information channels” & then processed in parallel by different cells at all levels



- Cornea(concentrate) & Lens(focuses on distance) focus light
- When you age, muscles around lens get less flexible
 - Leads to a type of farsightedness
- Farsightedness is when eye refracts too late
- Myopia (nearsightedness) eye refracts too early
- Retina turns light into electrochemical signals
 - Made up of photoreceptors & every part of the eye helps them:
 1. Stay alive
 2. Do their job
 - Fovea is the retina's center(best vision)
- Pupil is the hole that changes size



- Photoreceptors:

1. Rods - light vs. dark(outside fovea)[rhodospin]

2. Cones - color(clumped in fovea)

- Disk membranes to help absorb light
- Cones come in three types(Red, green, & blue)
 - Absorption spectrum: light photoreceptors can absorb

- Light causes hyperpolarization(changes ospin's shape)

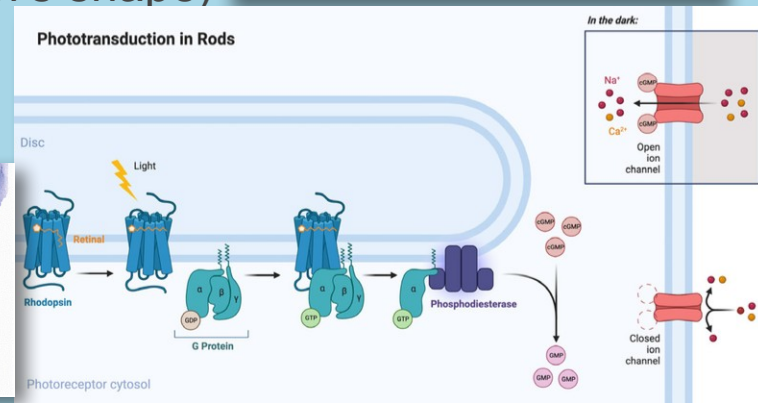
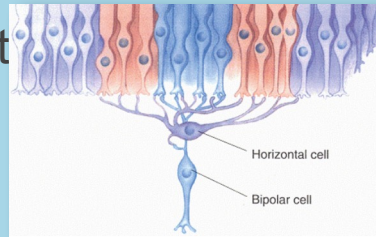
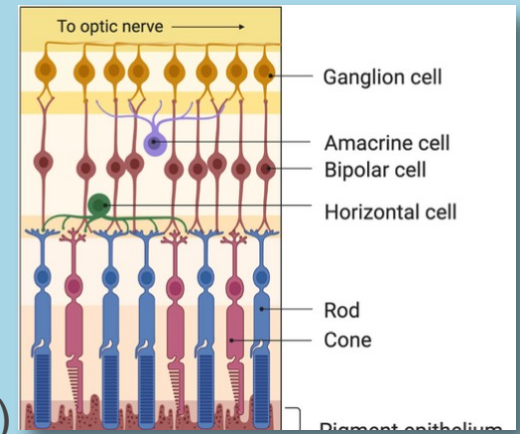
- Signals are sent because of change in stimulus
- The changed ospin can activate a GP
 - Blocks an AP

- In the dark, signals are sent

- Bipolar cells
- -> Retinal Ganglion cells
- -> Optic Nerve(blind spot)

- Acramine cells connect bipolar to ganglion

- Horizontal cells connect other photoreceptors



- **Lateral inhibition:** how much excited neurons can reduce activity of neighbors

- When a photoreceptor depolarizes a horizontal cell, which
- To sharpen edges & make contrast
- Smaller receptive field -> sharper vision

- Retinofugal pathway(seeing pictures):

1. Optic Nerve

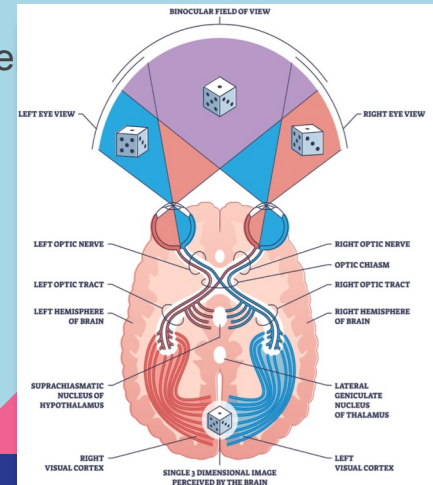
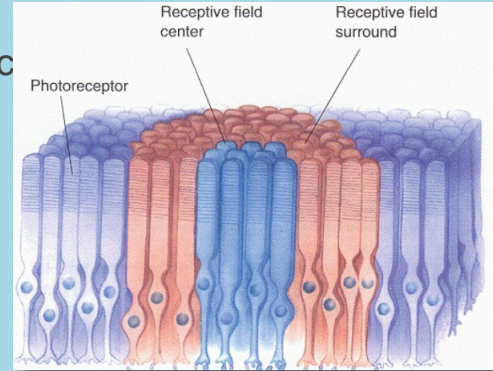
2. Thalamus(Lateral geniculate nucleus[LGN])

- LGN processes visual input

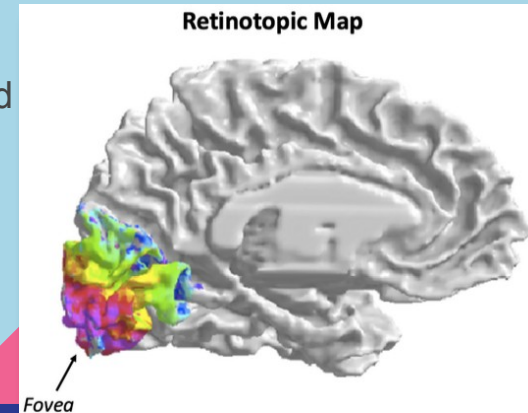
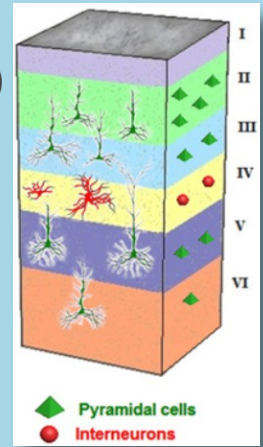
- uses inputs from other brain parts to decide how to see from the eye

1. Occipital lobe

- Some axons cross over(optic chiasm)[the greek letter “chi”]
 - How binocular vision works
- Other paths tell the eyes how to respond



- LGN sends signals to The Primary visual cortex(V1[striate cortex])
 - Initial processing of visual input
- V1 has many layers, most input goes to layer 4(most complex)
 - Layer 4C(a sublayer of layer 4) has stellate neurons
 - Every other layer has pyramidal cells(long dendrite & long axon)
 - Once info moves beyond 4C, information from both eyes starts to mix
- Retinotopic map of the visual world that is on the retina is in V1
- The map is backwards & upside down from reality, center is bigger
 - More neurons process objects that are in the center than object in the peripheral view
- Visual input moves to the extrastriate cortex
 - Process the signal further
 - Path of info from V1 to other areas depends on the info processed



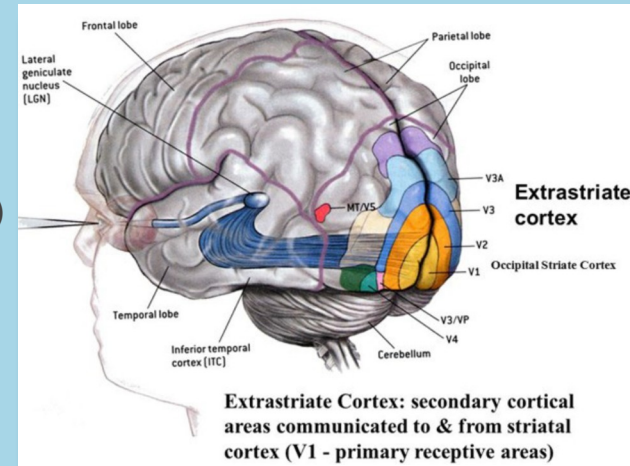
- Mostly, info about location & motion goes to top of occipital & parietal lobes(Dorsal Extrastriate Pathway)
 - Processes the “where” of sight
 - Damage leads to akinetopsia(motion blindness)
- Info about color, shape, & form is sent to the bottom parts of the occipital & temporal lobes(Ventral Extrastriate Pathway)

- Process recognition of:

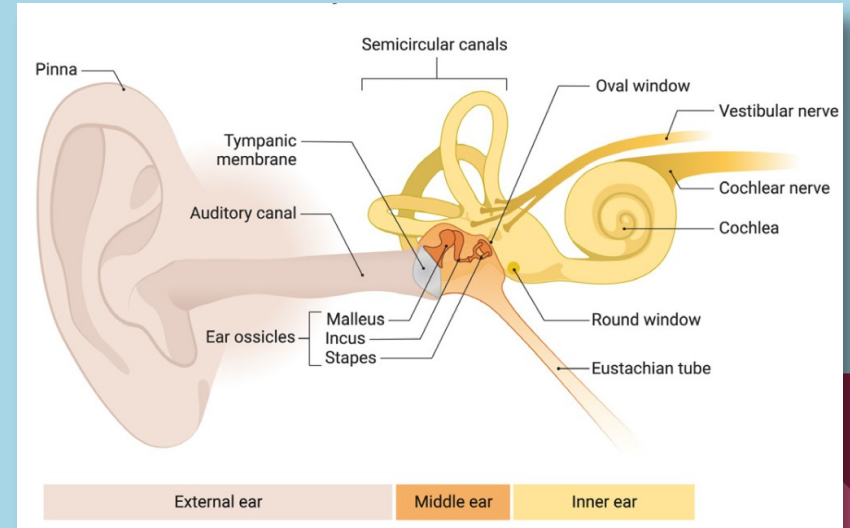
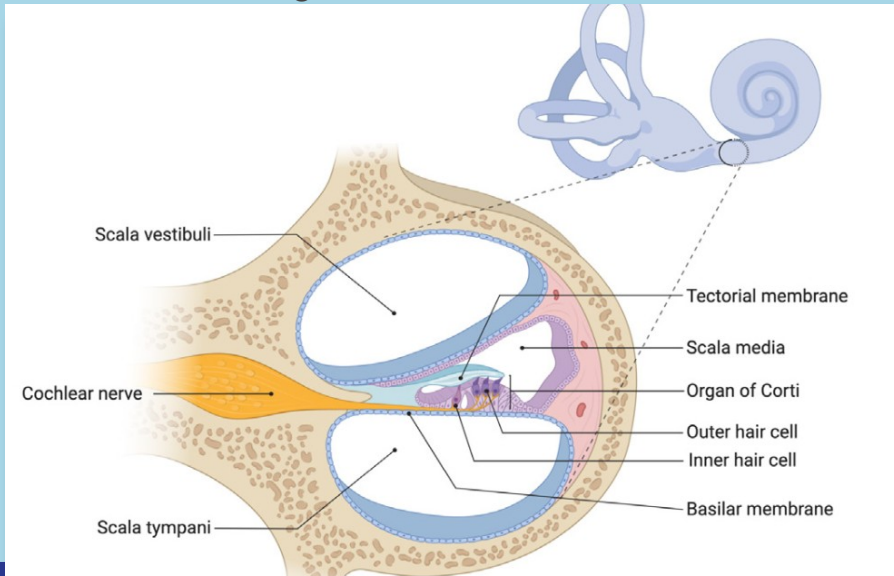
1. Objects
2. Faces
3. Symbols
4. Colors
5. Text



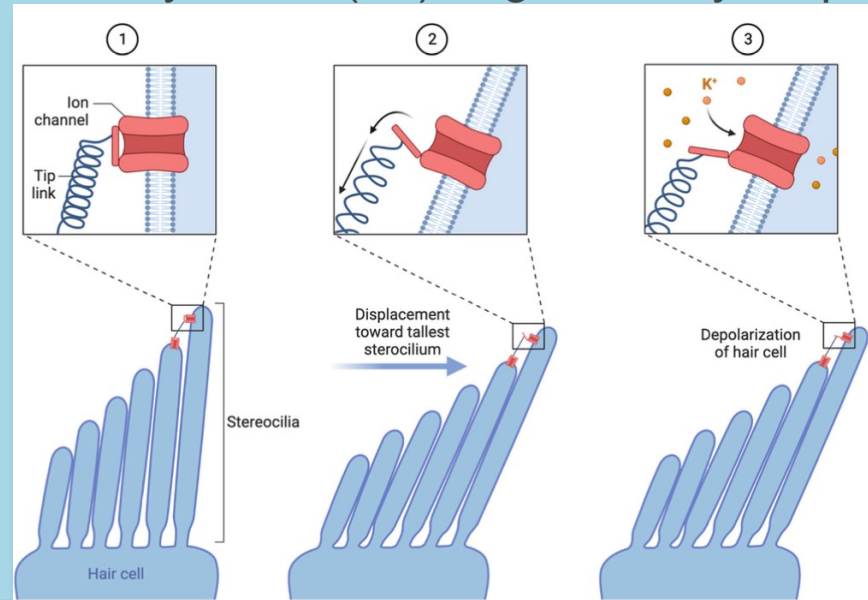
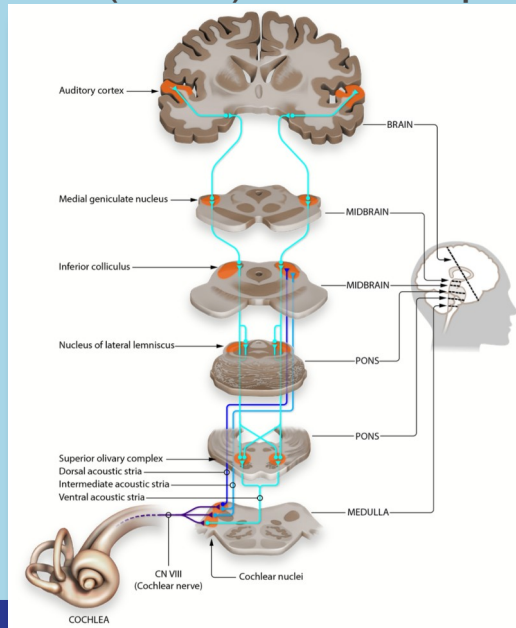
- Damage leads to visual agnosia(deficit in object recognition)
 - Most famous is prosopagnosia(face blindness)



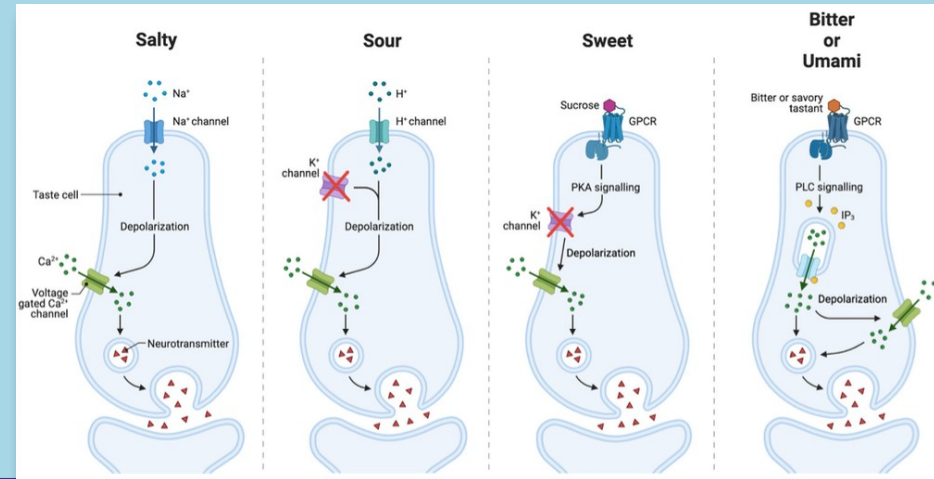
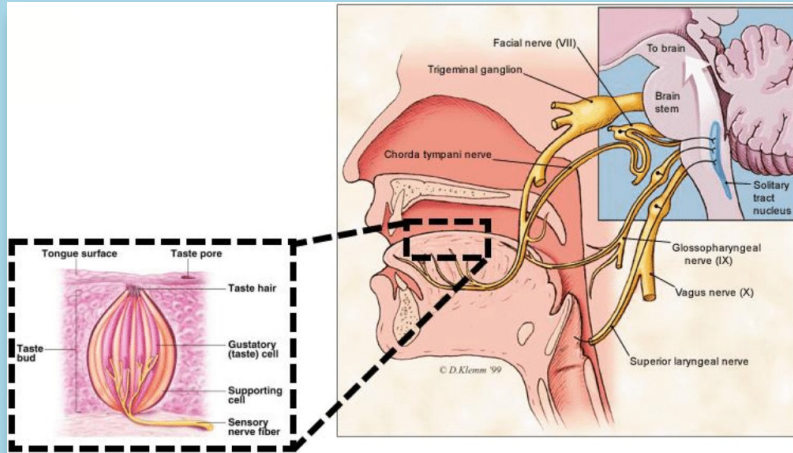
- Sound involves changes in air pressure, detected through the ear structures: pinnae, auditory canal, & tympanic membrane
- Auditory system converts sound into mechanical movements via ossicles (malleus, incus, stapes)
 - turns mechanical movements into fluid waves in the cochlea
 - Stapes vibrating at the oval window initiate fluid movement in the cochlea(snail), activating the Organ of Corti



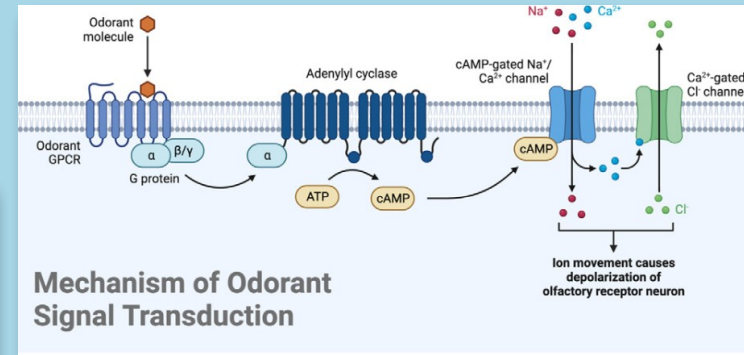
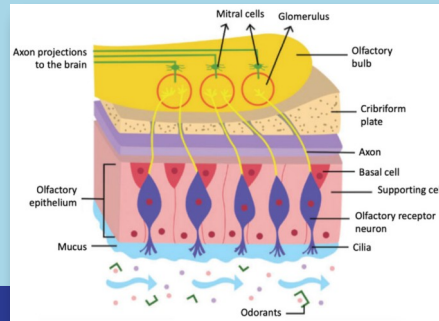
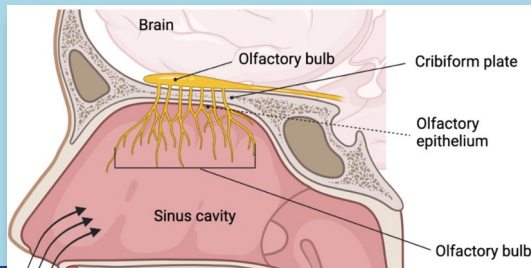
- Inner hair cells turn vibrations into electrical signals, while outer hair cells amplify volume
- Auditory pathway involves complex networks of brainstem nuclei
 - Brainstem calculates sound origin using intensity differences & time delays between ears
- Signals from both ears mix early in the pathway & reach the medial geniculate nucleus (MGN), then the primary auditory cortex (A1), organized by frequency



- Chemical senses include taste & smell
 - Identifies basic tastes (sweet, sour, bitter, salty, umami) through specialized receptors & transduction pathways
 - Taste signals travel via cranial nerves to the brain
 - Salt & sour: ions enter the taste cell & depolarize it
 - Sweet & bitter/umami: sucrose/tastant bind to GPCR, which depolarizes the cell



- Smell involves a G protein-coupled signaling cascade in olfactory neurons
- Olfactory receptor neuron axons signal glomeruli after passing through the cribriform plate
- People have about 2,000 glomeruli
 - axons are grouped by specific receptors
- Olfactory signals bypass the thalamus, directly reaching the brain's cortex, hippocampus, & amygdala
- **Population Coding:** where multiple receptor types contribute to the perception of a single sensation
 - preserves the ability to taste or smell
 - made by patterns of neuron activation



- Skin Layers:

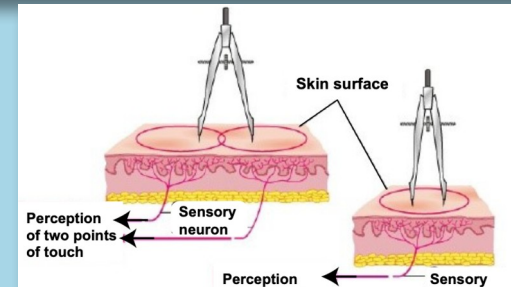
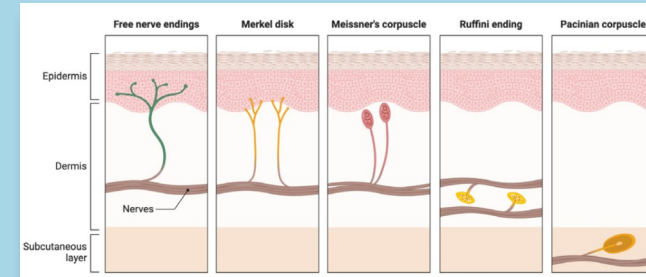
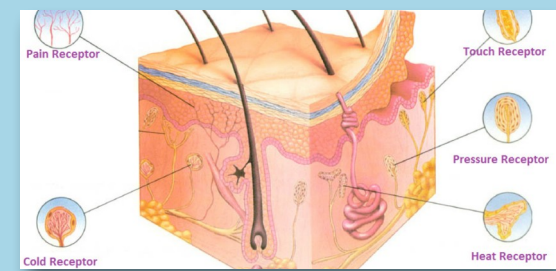
- Epidermis: Outer barrier
- Dermis: Contains blood vessels, glands, hair follicles
- Subcutaneous: Fatty layer w/ sensory neuron axons

- Touch Receptors:

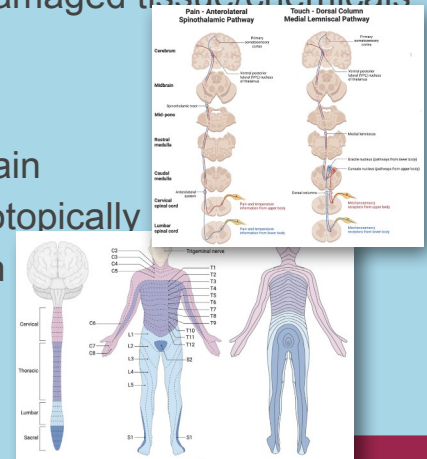
1. Meissner's corpuscles - motion & grip
 2. Pacinian corpuscles - vibrations
 3. Merkel disks - shape/texture
 4. Ruffini ending - skin stretch & tangential force
- Specialized for touch, pressure, vibration; travel fast
 - Pacinian & Meissner's corpuscles adapt quickly to stimuli
 - Merkel disks & Ruffini endings maintain response during stimulus

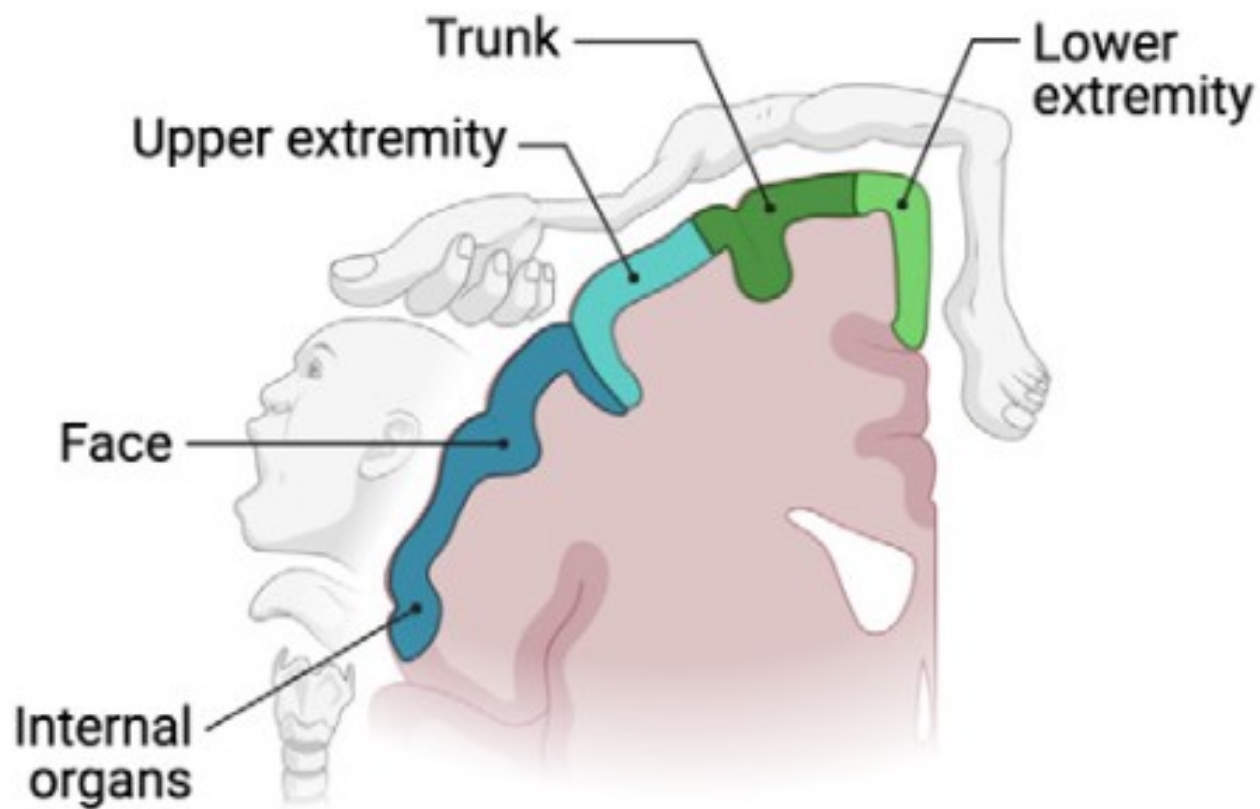
- Receptive Fields:

- Surface-close receptors have precise receptive fields
- Deep receptors like Ruffini & Pacinian have larger, less precise fields
- **Two-point discrimination:** Acuity of the skin
 - highest in fingertips, lips, tongue

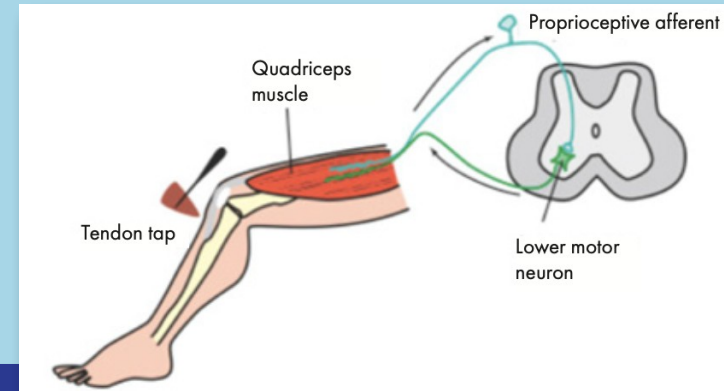
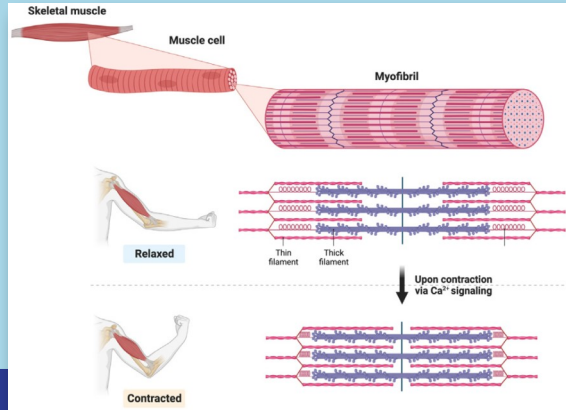


- **Dermatomes:** skin areas innervated by spinal cord segments; useful for diagnosing conditions like shingles
- Temperature & Pain Receptors:
 - Free nerve endings detect temperature & pain(different receptors for non-damaging & scary temp.)
 - Pain signals processed through nociception; can be influenced by damaged tissue/chemicals like capsaicin
- Pathways:
 - Signals from receptors travel via spinal cord/cranial nerves to the brain
 - Primary somatosensory cortex processes signals, organized somatotopically
- Somatosensory Cortex Mapping:
 - Spatial arrangement reflects body part representation(not proportional to body size)
 - Sensory homunculus visualizes cortical representation(based on receptor density)





- **Muscle Activation:**
 - Lower motor neurons stimulate muscles w/ acetylcholine at the neuromuscular junction, causing muscle contraction through a calcium-triggered process
- **Motor Neurons & Reflexes:**
 - Lower motor neurons, located in the spinal cord or brainstem, connect directly to muscles
 - Reflex circuits enable immediate motor responses w/o brain input, shown by the knee-jerk reflex
- **Voluntary Movement & the Motor Cortex:**
 - The primary motor cortex (M1) plans & coordinates voluntary movements, w/ input from supplementary & premotor cortices
 - Movement planning involves selecting muscles & limbs to achieve behavioral goals

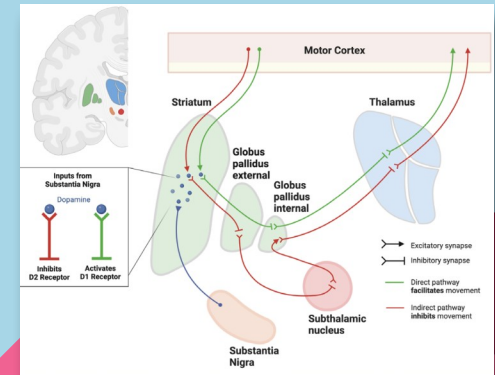
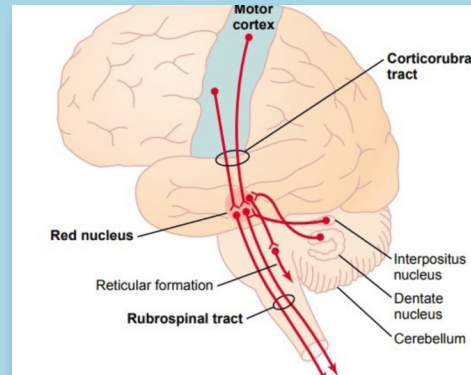
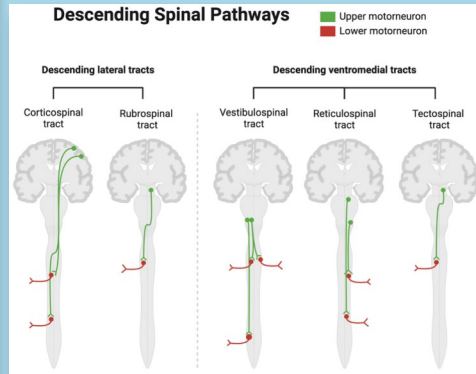
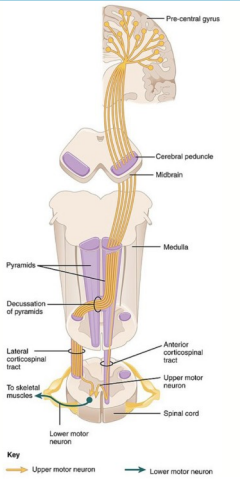


• Descending Motor Pathways:

- Voluntary movements are controlled by the lateral pathways (corticospinal & rubrospinal tracts), facilitating fine limb movements
- Ventromedial pathways (vestibulospinal, tectospinal, reticulospinal tracts) manage posture & balance, targeting core & trunk muscles

• Basal Ganglia & Motor Control:

- The basal ganglia regulate initiation of movement, w/ damage leading to motor impairments like those seen in Parkinson's & Huntington's diseases
- Consist of complex circuits, including the direct pathway (facilitates movement) & the indirect pathway (inhibits unwanted movement), influenced by dopamine from the substantia nigra





Synaptic Plasticity and Memory

Plasticity: The brain's ability to change & adapt to new information → **Synaptic Plasticity** (change that occurs at synapses)

Dr. Donald Hebb first proposed Hebbian Theory in 1949

- Proposes that cells that as certain connections are used more frequently, they become stronger & the neurons more tightly linked (“cells that fire together, wire together”).
 - This **synaptic strength** can change depending on environment & modifications in input.
 - Likely mechanism for memory encoding & storage (making heavily researched).

EXPERIENCE-BASED CORTICAL CHANGES

The inputs to the visual cortex from the thalamus are segregated by eye, These separate inputs target separate neurons in layer 4C, giving rise to **ocular dominance columns**, (stripes of tissue across 4C that get input from 1 eye only, ODCs).

- The concept introduced by David Hubel & Torsten Wiesel in the 1960s. Used cats looking at light on a screen to see the neurons in 4C fired action potentials in response to input from one eye or the other, but not both.
 - Performed **monocular deprivation** experiments (one eye is sealed shut soon after birth so it doesn't develop). Found that the ODCs in the sealed eye shrank away while the open eye's columns filled the space.
 - Even short periods of monoc. dep. can impact the eye.

Classic experiments like these led to the idea of **critical periods** (time in which plasticity is high & the development is influenced based on the environmental inputs they receive) They can exist for all senses.

- Short & usually take place during early development, first few weeks to months of life.

The somatosensory cortex can also change due to plasticity. it (often represented as a homunculus) permanently change

- With whiskers, touch from each one is represented in barrel). The entire area of whisker representation loo

Some aspects of cortical plasticity in the barrel field are spec newborn rat, the barrels will never form for those whiskers i

- Thus, the expansion for these whisker representation

This very same concept of dedicated cerebral sections being overtaken (cortical reorganization) applies to primates as well.

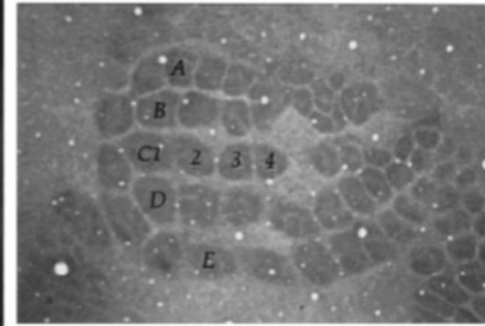
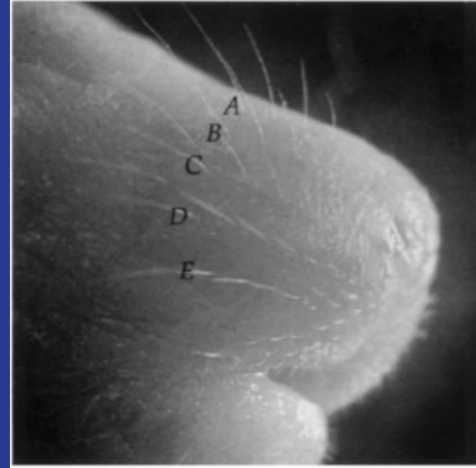
This sensory adaptation works best in children & can go aw

- Ex: **Phantom limb syndrome** (a patient who lost that missing part of the body). Often result of stimul missing part.

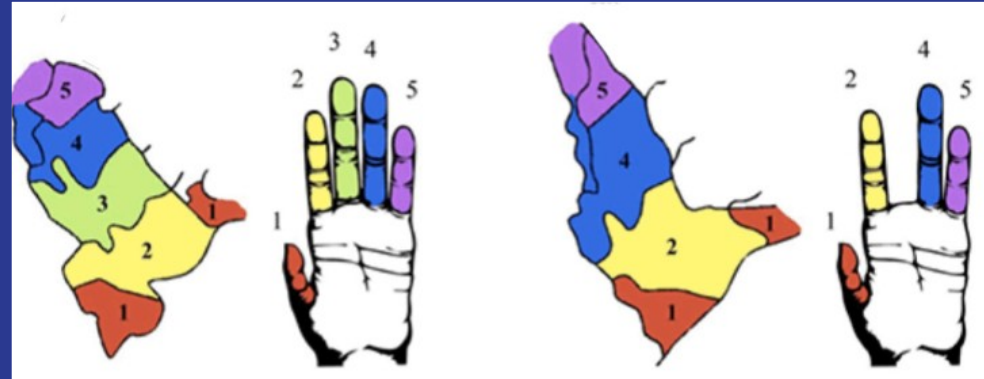
STRENGTHENING AN

Although the brain becomes less plastic via aging, it never s info from its experiences. Both these processes—learning &

- Spanish neuroscientist Santiago Ramon y Cajal first speculated in 1894 that learning requires new connections between neurons.



Whiskers on a 1 day old rat pup can be seen on the left, and the corresponding barrels can be seen on the right.



In the 1960s scientists like Eric Kandel used new technology to study memory circuits in simple animals, like the sea slug Aplysia californica. These animals can learn very basic things, such as associations.

- By repeatedly touching the slugs gills (causing them to retract), one gets **habituation** (learning to ignore a stimulus that lacks meaning). Is a form of learning.

Kandel earned Nobel Prize by showing this change in behavior is a change in the strength of the synapse between the sensory neuron & motor neuron. The slug fired the same type of action potential, but it had a lesser impact on the postsynaptic cell.

If researchers used a painful electrical shock, the slug would learn the touch is meaningful very meaningful.

- This causes **sensitization** (strengthening of behavioral response, via ↑Neurotransmitters, ↑Synapse surface area).

Serotonin from the interneuron binds to GPCRs in the terminal of the siphon sensory neuron, which stimulates the production of cAMP, which activates protein kinase A (PKA). PKA phosphorylates several proteins.. This reduces the ability K⁺ channels' ability to open during an action potential, prolonging the presynaptic action potential in the terminal, allowing more Ca²⁺ channels in the terminal to open, extending exocytosis & increasing the amount of the neurotransmitter glutamate that is released onto the motor neuron.

Similar patterns of synaptic strengthening & synaptic weakening encode learning & memory in humans too. There are other possible mechanisms for strengthening/weakening synapses, including changing the postsynaptic receptors.

Researchers can measure if the strength of a synapse changes due to experience w/ **electrophysiological recordings**. Done by recording the membrane potential of the presynaptic & postsynaptic neuron before & after a “learning” event.

Long-term potentiation (LTP) (strengthening of a synapse based on changes in the postsynaptic receptors) is commonly studied. Can potentially last for an entire lifetime. Linked to memory storage, especially in the hippocampus, responsible for **consolidating** memories (stabilizing them for long-term storage).

LTP happens when presynaptic & postsynaptic cells are depolarized at once. When the presynaptic is depolarized it releases glutamate. The postsynaptic has two different receptors: **α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptor**, & **N-methyl-D-aspartate (NMDA) receptor**.

- AMPA receptors: ionotropic & allow Na to depolarize the cell & make it more positive.
- NMDA receptors: binds to glutamate, usually blocked by Mg⁺. If enough depolarization by Na influx (via AMPA receptors), repelling the Mg⁺, allowing ions in. NMDA receptors allows sodium & calcium into the postsynaptic cell.

Calcium influx triggers **calcium calmodulin-dependent protein kinase (CaMKII)**, a messenger enzyme that guides new AMPA receptors to be inserted to ultimately strengthen postsynaptic response to the same amount of NT release. Phosphorylates AMPA receptors, making them work more efficiently. Can also lead to changes to the dendritic spine shape. Phosphorylates **CREB**, a transcription factor, critical for producing new proteins necessary to maintaining these structural changes. All these changes produce the LTP state.

Other signals can cause AMPA receptors to be removed from the postsynaptic membrane, weakening the synaptic connection, **long-term depression (LTD)**. While this is the most well-studied.

MEMORY SYSTEMS

Memory requires not just the encoding of pure facts, but also of context. The brain's memory system tends to focus its resources on important memories, ones linked to a strong emotional response or items that regularly occur. The hippocampus plays a key role in picking which memories to keep & storing those memories properly.

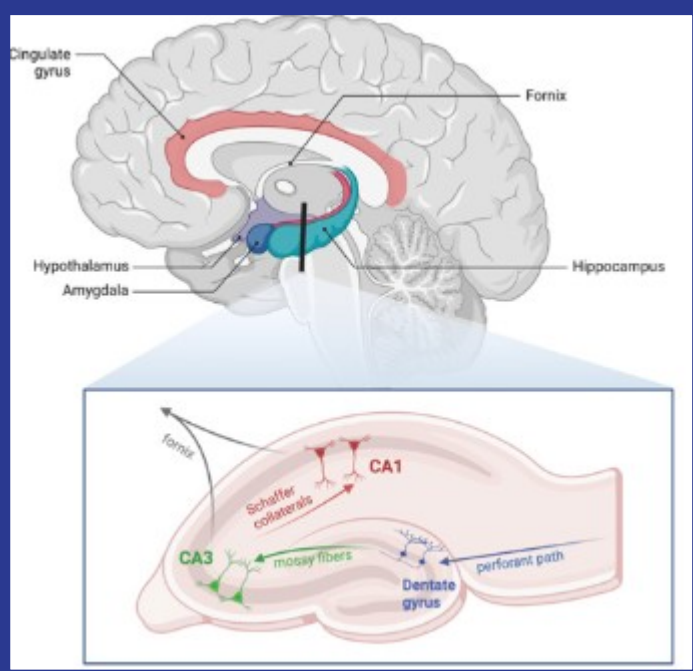
The hippocampus & related regions compile information from the entorhinal cortex. The hippocampus receives connections from the **limbic**/emotional regions (amygdala, hypothalamus).

- H.M., a young man w/ severe epilepsy, had surgery to remove his left & right hippocampus, could easily remember recent events but could not form new memories.

Most of the input to the hippocampus comes from the entorhinal cortex. It receives perceptual & sensory inputs before sending them to the hippocampus.

The entorhinal cortex cells send their axons to the hippocampus. One of the few locations in the human brain where information from all sensory modalities is processed in this circuit where information from all sensory modalities is processed together.

Small neurons in the dentate gyrus called granule cells send their axons (**mossy fibers**) to the hippocampus' main part, Ammon's Horn (Latin: *Cornu Ammonis*). The fibers synapse in a subsection, Cornu Ammonis 3 (CA3), on large neurons called pyramidal cells. These neurons in turn project mainly to other pyramidal cells in the CA1 subregion via axons called **Schaffer collaterals**, which are crucial for the consolidation of short-term memory into long-term memory.



It receives information from the entorhinal cortex. Also inputs from the limbic system, which are related to its content.

As a result of losing both the left & right hippocampus, H.M. could not produce new long-term memories.

Like most other regions, & helps process information. It is more simplistic, yet no less critical.

In the hippocampus, the **dentate gyrus**, one of the few locations in the human brain where information from all sensory modalities is processed together, can be seen in adults. The first region where information from all sensory modalities is processed together, can be seen in adults. The first region where information from all sensory modalities is processed together, can be seen in adults.

The axons of the pyramidal cells project out of the hippocampus, forming the **fornix** (brings info from the hippocampus to many deep brain regions). Some output from the hippocampus returns to the entorhinal cortex, forming a loop. Plasticity & LTP at each of these synapses appears to be critical for different types of long-term memory encoding & consolidation

Short-term memory is short lived requires conscious attention. Lets you retain a piece of info for a few minutes & doesn't last long after you stop focusing on it, like repeating a list of items that has just been read to you. People can retain 5-9 items. Prefrontal cortex, especially the top regions & those on the side, seems to be critical for maintaining this type of memory.

Long-term memory results from 3 main processes: encoding, storage, & retrieval of info. Encoding: learning/acquiring new info. Storage: filing away this new info even when you aren't paying attention to it or thinking about it consciously. Retrieval: recall of info. Can be explicit & implicit memory, which can be divided into various subtypes.

- **Explicit memory**: any memory that can be consciously recalled. Explicit memories can be **semantic memories**, meaning facts without any context/emotions attached or items you would study for a test. Can be **episodic**, memories related to experiences or “episodes” in your life (10th birthday party). Explicit memory must be consolidated by the hippocampus to become table for long-term storage, but implicit memory does not use the hippocampus.
- **Implicit memory**: unconscious & unrehearsed memory. Are usually slower to encode, take more effort to learn, stick around for a lifetime. Doesn't require the hippocampus.
 - **Procedural memory**: Learned motor skills. Involves motor regions of the brain like the cerebellum. Habits are also a form of implicit memory, relying on the basal ganglia.

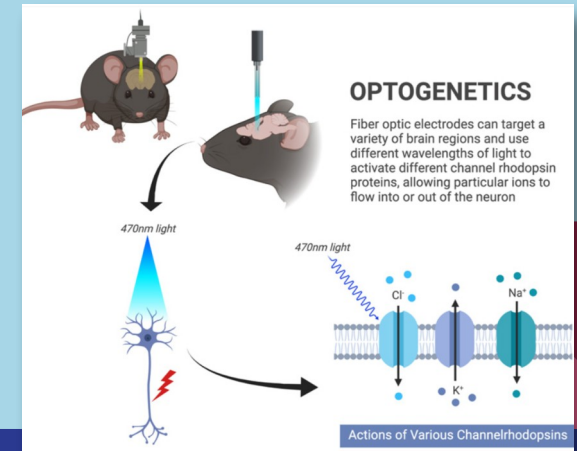
- Implicit memories can also result from **priming**, which occurs when exposure to one stimulus influences your memory.
 - Ex: You'd more successfully remember words from a list that were related to health medicine if you were first shown pictures of a doctor's office.
- Emotional associations are another form of implicit memory. **Classical conditioning** is a common example.
 - Involves getting a subject to form an association between a meaningful stimulus (a tasty treat) & a neutral item (the sound of a bell). After repeated exposures to the meaningful stimulus w/ the neutral item, they'll closely associate the two even if they're unaware of the association. Like Kandel and the *Aplysia californica*. In mammals, conditioning relies on limbic regions (amygdala & cerebellum).

WE'RE ALMOST DONE WOO-HOO!!!

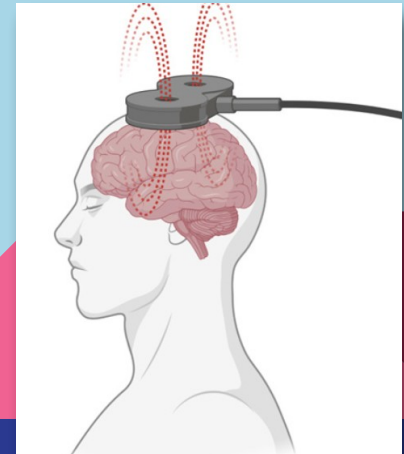
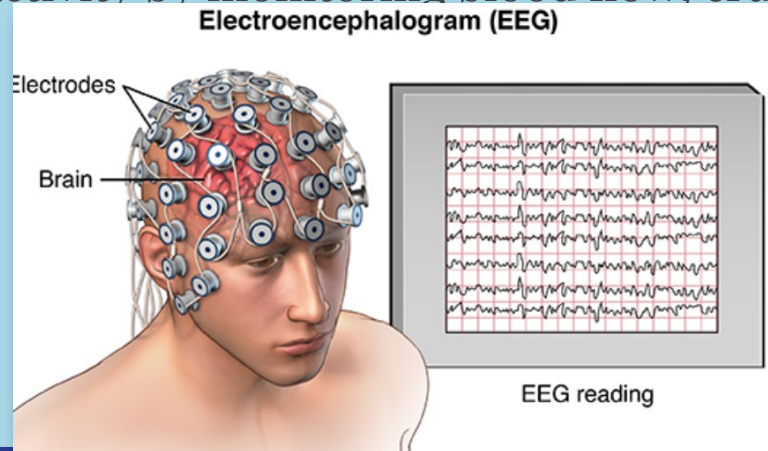


Technology and Neuroscience

- **Neuroscience & Technology Development**
 - The field has rapidly advanced w/ technological innovations, including the Brain Research through Advancing Innovative Neurotechnologies (BRAIN) Initiative launched in 2013 by President Obama
 - Developments include the Human Connectome Project & advanced microscopy techniques
- **Methods of Perturbing the Brain:**
 - Early studies involved lesioning or removing brain regions to study function, while modern approaches use noninvasive techniques like electrical stimulation, TMS, & optogenetics
- **Electrical Stimulation & Optogenetics:**
 - Electrical stimulation, used since the 19th century, & optogenetics, a more recent technique, allow targeted activation or inhibition of neurons



- **Transcranial Magnetic Stimulation (TMS):**
 - TMS noninvasively modulates brain activity, used both in research & clinically to treat depression
- **Recording from the Brain:**
 - Techniques range from invasive single-unit recordings to noninvasive EEGs, providing insights into brain activity
- **Magnetic Resonance Imaging (MRI) & Functional MRI (fMRI):**
 1. MRI provides detailed internal images
 2. fMRI tracks brain activity by monitoring blood flow, crucial for cognitive neuroscience



- **Brain-Machine Interface (BMI):**
 - BMIs translate neural activity into commands to control devices, offering potential benefits for people with disabilities & advancing human-computer interactions
- **Computational Neuroscience:**
 - **Computational neuronal modeling:** Mathematical models & simulations to understand brain functions, from individual neurons to entire networks
- **Artificial Intelligence (AI) & Computational Psychiatry:**
 - AI benefits from neuroscience insights to develop neural network-based learning systems
 - Computational psychiatry aims to use computational models to improve psychiatric diagnosis & treatment

